Sohum Mehta

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Education

University of Pennsylvania Philadelphia, PA

Computer Science and Cognitive Science (BA); Minor: Statistics and Data Science (Wharton)

- **Coursework**: CIS 1200 Programming Languages & Techniques; MATH 1410 Calculus II (upcoming)

Illinois Mathematics and Science Academy Aurora, IL

High School Diploma | GPA: 3.98/4.00 (Unweighted) | Coursework: ML, NeuroAI, Differential Equations

Y Combinator Startup School '26 | *Became a founder in San Francisco; research presentations & hackathons*

Research Experience

Google Quantum AI | *Explainable AI Research Intern, Variational Quantum Algorithms (VQE/VQC)*

- Built **parameter-shift saliency** for VQCs (∼232× confidence-drop gain), exposing a one-pixel shortcut
- Accepted at **IEEE/ACM AI-ML Systems**, **NeurIPS 2025 ML4PS** (sole author), and **YQIS 2025**

Carnegie Mellon University | *Computational Neurolinguistics Student Researcher*

- Modeled lexical retrieval (**EEG** + RT, ∼20K points): **high-frequency words mask conceptual priming**
- **National JSHS Finalist** (top ∼3%, $1,000); oral presentation accepted at **Springer Brain Informatics**

Winter Lab, Northwestern (Feinberg) | *Bioinformatics Research Intern*

- Engineered a lab-wide **single-cell RNA-seq web app** in R; cut time-to-first-plot from minutes to seconds
- Led biomarker discovery validated by immunologists; co-authored the resulting **ACM-BCB 2025** paper

Penn Undergraduate Biotech Society (Wharton) | *AI Committee, Engram Project*

- Adapting Arc Institute's **Stack** foundation model (**149M cells**) to take bulk, multimodal, non-human references
- Inaugural AI Committee member; research-based client project directly supporting Engram's operations

Selected Publications and Presentations

TabFaith: Benchmarking and Improving Structural Faithfulness in LLM Table Summarization ↗ — ACL 2026 SURGeLLM (Oral, top 6 of 90+)

Energy Landscapes of Truthfulness in LLM Attention ↗ — ICLR 2026 NFAM

QCxAI: Parameter-Shift Saliency for Variational Quantum Classifiers ↗ — NeurIPS 2025 ML4PS

Prompt Genotyping: Quantifying the Evaluation Gap Between Synthetic Benchmarks and Real LLM Performance ↗ — NeurIPS 2025 LLM Evaluation

Cryptographic Fingerprinting for Medical AI ↗ — NeurIPS 2025 Lock-LLM

Cell-Type-Aware Single-Cell ML Reveals Distinct Biomarkers for Sarcoidosis Prognosis and CBD Differentiation ↗ — ACM-BCB 2025

Startups & Leadership

Foto AI | *Co-Founder · Electron, React, TypeScript, Swift, Apple Vision, Claude API* *foto-ai.store*

- Building a macOS agent that turns screenshots into calendar events, reminders and saved places in one click
- On-device OCR, LLM intent parsing, local calendar checks, verified writes with undo, injection-safe
- Secured **$100K in in-kind support from Y Combinator** after pitching YC partner Jared Friedman
- Lead operations in the **WUEC Incubator**, Penn's largest entrepreneurship club (**2,100+** members and alumni)

Wedge (YC S25) | *Internal Project Fellow, Software Engineering & Research*

- Only high-school operator selected for this **YC-backed fellowship (<0.7% chosen)**, with undergrad/grad teams
- Led applied-AI and workflow-automation work for **major hospital systems**; peers billed up to **$180/hr**

Promptonium | *Prompt-genotyping platform · Next.js, TypeScript, LLM APIs* *promptonium.vercel.app*

- Prompt genotyping research presented at **NeurIPS 2025** and **EMNLP 2025 BlackboxNLP**
- **Microsoft and OpenAI-supported ($2K)**; cited by Carnegie Mellon and Microsoft; used by PhD researchers

PennApps | *Board Member, Development & Marketing Teams*

- Organizing **PennApps XXVII**, the nation's **first student-run college hackathon** (founded 2009)
- One of only **two** board members on both teams; PennApps has drawn **2,000+ hackers** in a year

Pay it Forward (PiF) Science | *Co-Founder & Co-President, 501(c)(3) Nonprofit*

- STEM workshops for **3,000+ students** with ∼70 advisors and 9 NGO partners; ∼**$120K/yr** Google Ads Grant

Teaching: **MathMates** math tutor, twice weekly, and **CSSP** mentor (Netter Center work-study); **Access Engineering** instructor in Penn Engineering's largest service club

Clubs: Quantum Computing Club at Penn; Grey Matters at Penn (author); Wharton Undergraduate Statistics Society

Technical Skills

Languages: Python, Java, OCaml, TypeScript, JavaScript, R, MATLAB, C#, HTML/CSS

Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, Hugging Face, Cirq, React, Next.js, Git, LLM APIs